A hybrid multi-criteria decision making technique to improve the quality of hotel industry services in Mashhad City

Ali Mohaghar

Faculty of Management,
University of Tehran,
Tehran, Iran
Email: amohaghar@ut.ac.ir

Roya Eghbal

Department of Business Administration and MBA,
Payame Noor University,
P.O. Box 19395-3697, Karaj, Iran
Email: roya.eghbal@yahoo.com

Seyed Hojjat Bazazzadeh*

Young Researchers and Elite Club,
West Tehran Branch,
Islamic Azad University,
Tehran, Iran
Email: Seyed.hb@gmail.com
*Corresponding author

Abstract: Service industry is one of the most lucrative industries in the 21st century. To survive and succeed in competitive markets, service organisations pay special attention to the quality of their services. In between, hotel industry, as one of the most important parts of booming tourism industry in each country, has always been considered. The goal of conducting this study is to identify and rank the factors affecting the improvement of the quality of hotel industry services. For this purpose, 21 components were identified reviewing research literature and experts' views; and finally, two FUZZY ANP and FUZZY TOPSIS methods prioritised using combination method. The conclusion showed that having new and up-to-date hotel equipments, having professional staff for hoteling services and respecting customers are the first priorities. Therefore, regarding the above factors, it is suggested to hotel managers, to work on advancing services quality presented by this industry.

Keywords: service quality; hotel industry; MCDM; F ANP; F TOPSIS.

Reference to this paper should be made as follows: Mohaghar, A., Eghbal, R. and Bazazzadeh, S.H. (2017) 'A hybrid multi-criteria decision making technique to improve the quality of hotel industry services in Mashhad City', *Int. J. Knowledge Management in Tourism and Hospitality*, Vol. 1, No. 2, pp.152–175.

Copyright © 2017 Inderscience Enterprises Ltd.

Biographical notes: Ali Mohaghar received his BSc in Industrial Engineering from Sharif University of Technology, Tehran, Iran, in 1980, MSc degree in System-Oriented Planning OR from Isfahan University of Technology, Isfahan, Iran, in 1987, and PhD degree in the field of Industrial Engineering, Tarbiat Modarres University, in 2001. His research focuses on management, strategic management, supply chain management, MCDM, inventory control, and system dynamics. He is currently with the Department of Industrial Engineering of Tehran University as a Full Professor.

Roya Eghbal received her BSc degree in Agricultural Sciences and Natural Resources University, Gorgan, Iran, in 2005 and MBA from Payame Noor University, Karaj, Iran, in 2012. Her research focuses on decision making in marketing.

Seyed Hojjat Bazazzadeh received his BSc degree in Business Management from Payame Noor University, Tehran, Iran, in 2012 and MSc degree in Business Management from Payame Noor University, Tehran, Iran, in 2014; His research focuses on decision making in marketing. He is currently with the Department of Business Management of Islamic Azad University as a PhD candidate in marketing management.

1 Introduction

Today for survival, all organisations need to compete in different areas such as services. Therefore, providing high quality services is necessary and leads an organisation to a sustainable competitive advantage. Experiences clearly indicate that service delivery improvement would lead to customer satisfaction (Yaya et al., 2014), and customer satisfaction results in increasing his/her loyalty (Izogo and Ogba, 2015) that the result of which will be more profitable for an organisation (Sattar et al., 2014). Service quality is an enduring construct that encompasses quality performance in all activities undertaken by management and employees (Sofi et al., 2016). Service quality is a function of the difference between service expected and customers' perceptions of the actual service delivered (Hapsari et al., 2016). The discussion of service quality in residential places is considered as one of the important and essential discussions. Lack of the specified standards for service quality in the field of residential industry has taken the possibility of exact definition for standard service quality from the owners and managers in this industry (Borkar and Koranne, 2014). Obviously, what causes to maintain and survive the hotel industry is lack of gaps and contradictions between the expectations of customers and the delivered services (Prince and Khaleq, 2013). Effective customer management is a vital issue for the success of hotel industry (Prakash et al., 2016). Since, the goal of all hotels is a thriving business in the current competitive conditions, the most successful hotels can attract permanent or so-called loyal customers and this will not be achieved except by keeping the customers satisfied which needs to provide high quality services (Siddiqi, 2011). The best way to provide high quality services in hotels is to recognise customers' needs, demands and interests; and awareness of their ideas, suggestions and criticisms (Amritharaj and Vembar, 2014). Iran with a history of 7,000 years and having potential tourism capacities has always been considered by tourism industry managers. According to the statistics of Universal Tourism Organization (UNWTO); by attracting

5.3 million foreign tourists in 2013, Iran has been placed at the fourth rank in the Middle East. In between, historical cities of Shiraz, Isfahan, Yazd and Mashhad have been the destinations of most foreign tourists. After Tehran city, Mashhad is the second metropolis of Iran. Because of the existence of the shrine of Imam Ali ibn Musa al-Reza, the eighth Imam of Shiites in the world, this city annually has more than 30 million domestic and more than 1 million foreign pilgrims (Alizadeh, 2011). According to the selection of Mashhad as a research scope of this study, the discussion about providing services in residential places especially hotels plays an important role in attracting more tourists and development of tourism industry in this city. This study has been conducted on foreign passengers in four and five-star hotels that have travelled during March 2015 to visit holy, pilgrimage and tourism places of Mashhad city. As mentioned before, in this study we are looking for identification and prioritisation of the factors affecting service quality improvement using MADM techniques. Multi-attribute decision making model (MADM) is a scientific analytical model to evaluate and prioritise a set of options based on multiple criteria (Peng and Tzeng, 2013). So, in the second part of this study, the best service quality improvement model of hotel industry is recognised by reviewing research literature and help of experts and specialists in hotel industry. In the third part, namely research methodology, the algorithms of FUZZY ANP and FUSSY TOPSIS methods are described. The fourth part is related to data analysis. In this part, the weight of each criterion is obtained using FUZZY ANP method and the priority of each parameter achieved using FUSSY TOPSIS method. In the fifth part, conclusion will be stated and suggestions offered.

2 Research literature

2.1 Service quality

Since 1980, in addition to production sector, the discussion of quality has been considered in all parts and activities of an organisation such as service delivery (Gummesson et al., 2012). Quality measurement is easily done in tangible products and goods but measurement of this item in relation to services is somewhat vague and complex (Parasuraman et al., 1985). Service properties usually become clear in a dual relation between staff and costumers in a place that the service would be presented (Lun et al., 2016; Paul et al., 2016). Service quality as a source for creating competitive advantage has attracted a lot of attentions (Nejati et al., 2009); it is an important and significant issue for trade organisations all over the world (Haksever et al., 2015). Moreover, it plays an important role in promotion of comprehensive quality management (Yu et al., 2015). In fact, service quality is customers' judgments based on their total experiences about a company; this judgment is achieved as a result of comparing customers' expectations with the performance conceived by them (Dubé, 2015) and has a strong and direct impact on customers' confidence toward the considered organisation (Karjaluoto et al., 2015). Organisations can take steps towards customer orientation by training, socialisation and accountability (Whedon et al., 2016). In the recent years, great efforts have been made by experts to define and measure the quality of service conceived by customers (Babakus and Inhofe, 2015) and many researchers have studied the factors affecting the quality of service and as a result consumers' satisfaction (Bojanic, 2015; Cronin Jr. and Taylor, 1992). Bansal and Taylor (2015a) consider the delivered service

quality as an important potential factor for customers' satisfaction, Aniley and Negi (2015) state that there is a significant statistical relationship between customers' satisfaction and service quality; and generally, service quality is obtained through satisfaction. Thus, service quality is very important to create and enhance customer's satisfaction and understanding; it generates a friendly relationship between purchaser and seller (Dedeoglu et al., 2015). Hopkins et al. (2015) state that, the attraction and attitude of seller are among the factors affecting customers' perception of service quality. Guesalaga and Pitta (2014) has stated that, although, there is a relationship between service quality and key outcomes of it such as customer's satisfaction and performance; but, people's perceptions of service quality is different among various cultures. The results of Dubé's (2015) studies indicate that service quality improvement of a company marginally impacts on customers' satisfaction; therefore, it can be cited as a key factor in competitive advantage (Yang et al., 2011). Dabholkar (2015) surveys the direct and indirect impacts of customer's participation in service quality delivery; and suggests that increasing customer's participation for special service delivery will lead to higher quality of the perceived service. Caruana et al. (2015b) emphasise on the existence of a significant relationship between service quality and customer's loyalty and believe that managers and marketers should examine the dimensions of service quality in relation to their customers and meet customers' expectations (Jayawardhena, 2015). However, many customer-based firms have problem in the process of recognition and evaluation of customers' preferences and have often a false perception of customers' demands and expectations (Cronin et al., 2000). Parasuraman, that is one of the most famous marketing faces specially in the field of service quality (Bansal and Taylor, 2015b); mentions the five main dimensions of tangibility, reliability, responsiveness, assurance and empathy as the most important measurement criteria of service quality improvement (Gremler, 2011).

2.2 Service quality in the hotel industry and the factors affecting on it

The studies indicate that one of the factors influencing on thriving tourism industry is to consider accommodation services and hoteling for foreign travellers (Ye et al., 2014). Presenting good services to the costumers is one of most important factors of differentiation between hotels in a competitive atmosphere (Chang et al., 2016). Kozak and Gürel (2015) in their studies concluded that the quality of service delivery in the hotels has always been very important. In this respect, hoteliers are always looking for identification of effective factors of high quality service delivery to maintain its competitive advantage (Molina-Azorín et al., 2015; Nasution, 2016). Dhar (2015) in his research stresses on the existence of a strong and effective relationship between education of hotel employees and the delivered service quality. High quality service delivery is considered as an important factor for returning passengers to the hotel (Rauch et al., 2015). The results of Al-Refaie's (2015) researches indicate that use of modern methods of human resources management and creating satisfaction and loyalty in the hotel staffs will result in delivery of high quality service to customers. Siavoshani et al. (2014) in a research that was conducted in a hotel in Mashhad City concluded that the quality of service has a positive effect on reputation of name of a hotel. Hanzaee and Mirvaisi (2013) have mentioned the positive effect of staffs' job satisfaction on delivery of high quality service to customers. Managers and staffs tendency to deliver high quality service is one of the factors affecting customers' satisfaction (Vala, 2014; Rahimi and Kozak, 2016). Sorayaei et al. (2014) in their research have mentioned the committed and

competent staffs in delivery of high quality service. Varesi et al. (2012) in a research on the situation of service delivery in the hotels of Isfahan concluded that the most important weakness of these hotels is lack of effective facilities and trained and skilled staffs (Chen et al., 2011). Liat et al. (2014) mentioned that the factors such as creating positive image in customer mind as well as his/her satisfaction and loyalty is important to improve the quality of service in the hotel industry. Attractiveness of physical facilities as well as the tranquility dominant on hotel environment has a deep impact on passengers' satisfaction (Mueller and Kaufmann, 2001). Modernity of hotel facilities along with its apparent attractiveness is another factor affecting the improvement of hotel service quality (Lin, 2014). On the other side, delivery of high quality service and goods appropriate with the culture of passengers from different countries and also the performance of hotel staffs help to create a friendly space and distinguish this hotel from other hotels (Huei and Easvaralingam, 2011). Markovic et al. (2013) in their study mentioned the modernity of hotel facilities, the cleanliness of hotel, dress up of staffs and personnel's interest in solving guests' problems as outstanding cases. Dress up of staffs is important to create positive attitude in passengers (Pijls and Groen, 2012; Wang et al., 2013). Therefore, it should be tried to dress up hotel staffs; because, this attracts more attention to passengers and guests (Reichel et al., 2014). Ali and Amin (2014) considered a good and attractive physical environment as a factor for frequent travels to a hotel. Full performance of their obligations to passengers is of other factors effective on the improvement of hotel service quality (Zientara et al., 2015). Lin (2014) considered staffs' tendency to deliver service to customers and the existence of modern facilities in a hotel among the factors of services quality delivered to passengers that influence their satisfaction. Sohrabi et al. (2012) considered the quantity and quality of service facilities in a hotel, geographical location of a hotel, for example, being in a good climate region, proper delivery of service by staffs and finally courteous staffs of hotel among the factors affecting passengers' satisfaction. Mohsin and Lengler (2015) consider climate conditions of an area where the hotel is located as well as comparison of cost against services that deliver to customers effective on selecting a hotel by passengers. Arrival of passengers to clean rooms with beautiful landscapes impacts on their positive evaluation of a hotel (Walker, 1995). Restaurant also must be ready for favourable entertainment of guests (Borkar and Koranne, 2014). And also, hotel should provide various services for passengers' entertainment (Blešić et al., 2011). Ariffin and Maghzi (2012) considered smile and warm meeting of staffs, staffs efforts in solving passengers' problems and finally staffs treatment with respect effective on attracting passengers to a hotel. Anca and Andreea (2013) mentioned the cleanliness of hotel, staffs' tendency to deliver services to customers and respect to customers' character as the most important factors affecting the improvement of service quality. Another significant problem is that a balance should be considered between number of staffs and their responsibilities in the hotels (Chen et al., 2014). Rao (2014) states that staffs' tendency to deliver timely and accurate services are critical in the success of hotel. Among the factors affecting customers' satisfaction and return them to a hotel are interest in solving their problems, focus on customers' demands and respect to their character (Chadha, 2015). Also, the confidence of customers to the staffs and feeling that they are interested in solving passengers' problems and do their duties with utmost courtesy and respect are other factors that effect on the improvement of service quality (Minh et al., 2015). The results of Rhee and Yang's (2015) studies indicate that the location of hotel, service and facilities of rooms, residential pricing per night and the capability of staffs for rapid delivery service to passengers are among the

reasons for customers' satisfaction or dissatisfaction of a hotel. Attallah (2015) stresses on staffs' politeness when meeting passengers and their full knowledge and awareness in responding to probable questions. Dortyol et al. (2014) mentioned the factors such as creating confident in passengers, ability and communication skills of employees, attractive design and physical environment, staffs' empathy with the passengers and also the quality of food as the most important factors of improving service quality in a hotel. Martínez and del Bosque (2013) considered staffs' interest and tendency in solving customers' problems as the most important factor of creating satisfaction and loyalty. Deng and Zuo (2014) and Sitawati and Mia (2015) have stated that rapid delivery service to passengers is important. Wang et al. (2012) believed that hotels should consider current and potential needs of passengers to maintain their competitive advantage. Security of passengers is another important factor that should be on the agenda of hoteliers. Because feeling of security leads to a longer stay in the hotel (Tsai et al., 2015; Li et al., 2015b). Based on the review of research literature and using the views of university professors and experts in the hotel industry, the most important criteria and factors affecting the improvement of hotel service quality were identified according to Table 1.

Table 1 The factors affecting the improvement of service quality in the hotel industry

Criteria	*No*	*Factors*	*References*
Considering physical factors	1	Modernity of hotel facilities	Varesi et al. (2012), Lin (2014), Sohrabi et al. (2012), Markovic et al. (2013), Anca and Andreea (2013), Borkar and Koranne (2014) and Rhee and Yang (2015)
	2	Attractiveness of physical facilities in the hotel	Mueller and Kaufmann (2001), Lin (2014), Ali and Amin (2014) and Dortyol et al. (2014)
	3	Dress up of staffs	Pijls and Groen (2012), Sohrabi et al. (2012), Markovic et al. (2013), Wang et al. (2013) and Reichel et al. (2014)
	4	Locating hotel in a good climate region	Sohrabi et al. (2012), Mohsin and Lengler (2015) and Rhee and Yang (2015)
	5	A balance between number of staffs and their responsibilities in the hotels	Chen et al. (2014)
	6	The cleanliness of hotel environment	Walker (1995), Anca and Andreea (2013) and Markovic et al. (2013)
Confidence	7	Full performance of obligations	Zientara et al. (2015)
	8	Proper delivery of service by staffs	Huei and Easvaralingam (2011), Sohrabi et al. (2012) and Dortyol et al. (2014)
	9	Delivery of high quality service at the time promised	Rao (2014)
	10	Security of passengers	Tsai et al. (2015) and Li et al. (2015b)

Table 1 The factors affecting the improvement of service quality in the hotel industry (continued)

Criteria	*No*	*Factors*	*References*
Responding	11	Staffs' tendency in providing fast and reliable services to customers	Lin (2014), Anca and Andreea (2013), Rao (2014) and Minh et al. (2015)
	12	Staffs' tendency in solving customers' problems	Ariffin and Maghzi (2012), Markovic et al. (2013), Chadha (2015) and Minh et al. (2015)
	13	Appropriate response time	Deng and Zuo (2014), Sitawati et al. (2015) and Rhee and Yang (2015)
	14	Staffs' focus on addressing the needs and demands of customers	Chadha (2015)
Warranty services	15	The ability and credibility of hotel in providing reliable and good services	Minh et al. (2015) and Attallah (2015)
	16	Having the qualified and skilled staffs for hotel service delivery	Varesi et al. (2012), Attallah (2015) and Dhar (2015)
	17	Customers' trust in the hotel staffs	Martínez and del Bosque (2013) and Minh et al. (2015)
Respect for human values and empathy	18	Staffs' respect to customers' character	Ariffin and Maghzi (2012), Anca and Andreea (2013) and Chadha (2015)
	19	Particular attention of staffs to cultural differences of customers	Huei and Easvaralingam (2011)
	20	Understanding customers' needs by hotel staffs	Wang et al. (2012) and Ren et al. (2015)
	21	Politeness and humility of the hotel staffs	Sohrabi et al. (2012), Attallah (2015) and Minh et al. (2015)

3 Methodology

Decision making is considered as one of the most important duties of managers of companies and organisations (Betsch and Haberstroh, 2014). In fact, decision making is selection of an option among various options to solve a problem (Tzeng and Huang, 2011; Maier et al., 2014). Looking at the decisions that are made in the real world, it can be understood that most of our decisions are the types of multiple criteria decisions (Tzeng and Huang, 2011). In multiple criteria decision making instead of a criterion for optimality measurement, two or more criteria are used to compare solutions (Wan and Li,

2013). Multiple criteria decision making models are divided into two categories of multiple objective decision making and multiple attribute decision making (Gupta and Raman, 2014). In multiple objective decision making (MODM), the best option should be designed based on the limitations of the system, different objectives as well as optimal values considered by decision maker for these goals (Chen and Chang, 2014); while, multiple attribute decision making (MADM) is used for the selection of superior option and generally for ranking of options (Zeng et al., 2014). The set of MADM methods are considered as useful tools for making difficult decisions. Since, the criteria affecting the improvement of service quality have not equal importance and on the other side they are dependent on each other; among the set of multiple criteria decision making methods, ANP method was selected to determine the weight of criteria and TOPSIS method for final ranking of the components. But because classical management science uses precise and definitive data to analyse issues and this approach does not reflect thinking of human in the real world; so, fuzzy management science is used (Li et al., 2015a; Zadeh et al., 2014). Lotfi and Fallahnejad (2010) introduced the fuzzy set for the first time. In this set, verbal variables are used to resolve ambiguities and insert decision makers' feelings (Salah and Moselhi, 2015). The values of verbal variables are words or sentences found in natural language, but they are not numbers (Gottwald, 2015). Verbal variables can be quantified by use of fuzzy set theory and used in calculations (Huang, 2012; Karwowski and Mital 2014).

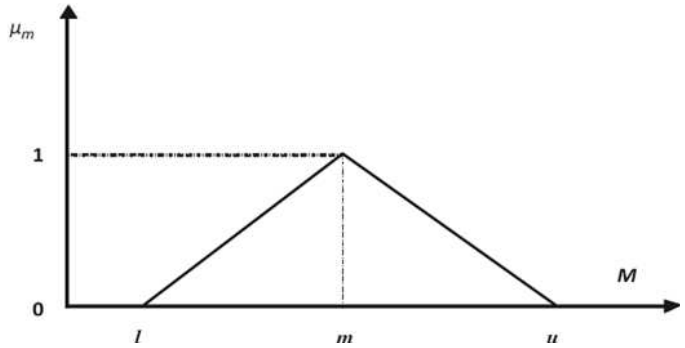
In classical set theory, an existence may be or may not be the member of a set Since, an existence is uncertainty in a fuzzy set, we should define the membership function, that is the fundamental of the fuzzy sets, for each existence in the set.

A fuzzy number is a particular fuzzy set $\tilde{A} = x, R \mid \mu_{\tilde{A}}(x),$ where x gets its values from the real line R_1: $-\infty < x < +\infty$ and its membership function $\mu_{\tilde{A}}(x)$ is a successive mapping from the closed interval and to it. Triangular and trapezoidal fuzzy numbers are the most generally used fuzzy numbers. Because of easy computation and features, triangular fuzzy numbers (TFNs) are often used in applications. TFNs are applied to represent the linguistic variables in this article. A TFN can be given as $\tilde{M} = (l, m, u),$, its membership function $\mu_{\tilde{M}}(x) : R \to [0, 1]$ is:

$$\mu_{\tilde{M}}(x) = \begin{cases} \dfrac{x-l}{m-l} & l \le x \le m \\ \dfrac{x-u}{m-u} & m \le x \le u \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where l and u are symbols of the lower and upper value of the support of $\tilde{M}$, respectively; and m is the mid-value of $\tilde{M}$. The parameters of l, m, and u describe a fuzzy number and imply the smallest possible value, the most promising value and the largest possible value, respectively. In Figure 1, TFN $\tilde{M}$ is shown 11 (Seçme et al., 2009).

Figure 1 Membership function of a TFN, $\tilde{M} = (l, m, u)$



If two positive TFNs $\tilde{A} = (l_1, m_1, u_1)$ and $\tilde{B} = (l_2, m_2, u_2)$ are defined, then we have:

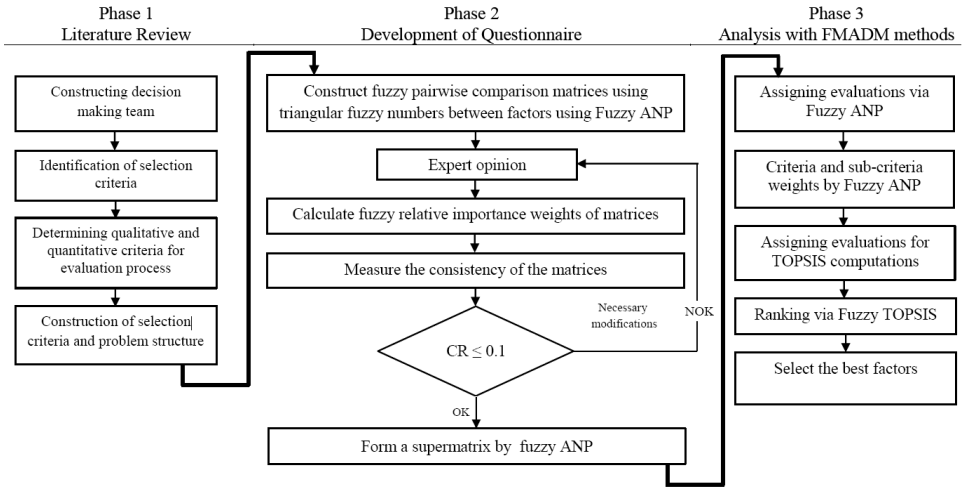
$$\tilde{A} \oplus \tilde{B} = (l_1, m_1, u_1) + (l_2, m_2, u_2) = (l_1 + l_2, m_1 + m_2, u_1 + u_2) \tag{2}$$

$$\tilde{A} \otimes \tilde{B} = (l_1, m_1, u_1) \times (l_2, m_2, u_2) = (l_1 \times l_2, m_1 \times m_2, u_1 \times u_2) \tag{3}$$

$$\tilde{A}^{-1} = (l_1, m_1, u_1) \approx \left(1/u_1, 1/m_1, 1/l_1 \right) \tag{4}$$

To analyse the data of this research, three main phases were defined according to Figure 2.

Figure 2 Analysis process of research data



As it is observed in Figure 2, reviewing research literature and using the views of university professors and experts in tourism industry; the most important criteria and factors affecting the improvement of the quality of hotel industry service were identified in phase 1. In phase 2, using paired-comparison matrix and Sevkli et al. (2012) fuzzy verbal expressions, expert survey questionnaire was provided to determine the

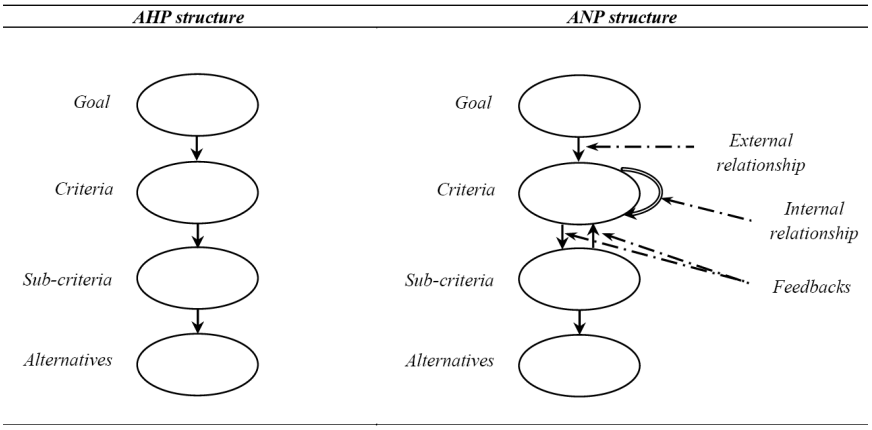
weight of criteria and distributed among ten experts. In continue the compatibility of paired-comparisons was analysed, using Gogus and Boucher's method (Gogus and Boucher, 1998). In the next phase, the second questionnaire in the format of Patil and Kant (2014) fuzzy quintuple Likert was distributed among 270 passengers of four and five-star hotels in Mashhad that 244 of them were returned. It should be noted that the validity of questionnaire data were confirmed by experts and Cronbach's alpha was calculated using SPSS software to ensure the reliability of data; the result was the number 8.0 indicating an acceptable reliability. In the third phase, MATLAB software was used; first, the weight of criteria was calculated using FUZZY ANP and next final prioritisation of components obtained using FUZZY TOPSIS. In continue the algorithm of each of FUZZY ANP and FUZZY TOPSIS methods will be described.

3.1 FANP

Due to disability of AHP method in considering the interdependencies between criteria and options, Saaty and Takizawa (1986) introduced the approach of network analysis process (Chang et al., 2015a). ANP is able to model the available correlations and feedbacks between elements affecting decision making and calculate all internal impacts of the elements affecting decision making (Chen, 2012). Although, paired-comparisons are used to prioritise the components in both AHP and ANP methods but there are differences between them. The first difference is that hierarchy analytic process is a special state of network analysis process; because, network analysis process considers the interdependence of cluster (interdependence) and middle clusters (external dependence) (Pastor-Ferrando et al., 2010).

The second difference is that the ANP has a nonlinear structure. Unlike the AHP model in which the decision maker considers a uni-directional hierarchical relationship between different levels of decisions (Aznar et al., 2010). ANP simply determines the relationships between criteria and options (Kilic et al., 2015). Figure 3 indicates the difference between AHP structure and ANP structure.

Figure 3 The difference between hierarchical structure and network structure



In ANP method, there is no need to strictly vertical hierarchical structure. In the networks, the elements of each class may impact on other or all elements. The

relationships are displayed in a network by arrows and where the direction of arrows is one-sided; the dependence is also one-sided (Kilic et al., 2015). The interdependence between the two classes is shown by a two-sided arrow and the interdependence between the elements of each class by ring shaped arrows (Chang et al., 2015a).

Although, experts use their own competence and capability to perform comparisons, but it should be noted that the network hierarchy analytic process may not fully reflect the style of human thinking (Tavana and Sodenkamp, 2010). According to the above descriptions, the use of fuzzy sets theory can eliminate this defect. So, in the algorithm proposed in this study, FANP method of Sevkli et al. was used for group decision making and determining the weight of each criterion. In this method, both input and output of FUZZY ANP technique are fuzzy and unlike classical fuzzy ANP, non-fuzzy structures (defuzzification) cannot be used; and paired-comparison matrix is completed between the criteria of each row, using TFNs. With this method, the values of parameters are obtained in the format of TFNs and calculated in fuzzy form. In the paired-comparison criteria, decision maker (expert) can use TFNs to determine degree of preference in options (Tadic et al., 2014). In classic ANP, the range of one to nine hours is used for paired-comparison performances. Although, this discrete spectrum is very simple and easy but does not include uncertainty and ambiguities related to an individual's understanding and judgment on degree of preference. In other words, the decision maker may not be able to express in comparison a certain number as degree of preference (Sevkli et al., 2012). That is why a range of 19 can be used for TFNs instead of a reasonable range of one to nine hours in classic ANP that is shown in Table 2.

Table 2 Definition of TFN-linguistic scale for importance

TFN	*Linguistic scale for importance*	*Triangular fuzzy scale*
$\tilde{1}$	Equally preferred	(1, 1, 1)
$\tilde{2}$	Equally to moderately preferred	(1, 3/2, 3/2)
$\tilde{3}$	Moderately preferred	(1, 2, 2)
$\tilde{4}$	Moderately to strongly preferred	(3, 7/2, 4)
$\tilde{5}$	Strongly preferred	(3, 4, 9/2)
$\tilde{6}$	Strongly to very strongly preferred	(3, 9/2, 5)
$\tilde{7}$	Very strongly preferred	(5, 11/2, 6)
$\tilde{8}$	Very strongly to extremely preferred	(5, 6, 7)
$\tilde{9}$	Extremely preferred	(5, 7, 9)

In order to evaluate decision maker's preferences, paired-comparison matrix is formed by use of TFNs (l, m, u) in Table 1. TFNs matrix ($m \times n$) can be shown as equation (5):

$$\tilde{A} = \begin{pmatrix} \left(a_{11}^{l}, a_{11}^{m}, a_{11}^{u}\right) & \left(a_{12}^{l}, a_{12}^{m}, a_{12}^{u}\right) & \cdots & \left(a_{1n}^{l}, a_{1n}^{m}, a_{1n}^{u}\right) \\ \left(a_{21}^{l}, a_{21}^{m}, a_{21}^{u}\right) & \left(a_{22}^{l}, a_{22}^{m}, a_{22}^{u}\right) & \cdots & \left(a_{2n}^{l}, a_{2n}^{m}, a_{2n}^{u}\right) \\ \vdots & \vdots & \ddots & \vdots \\ \left(a_{m1}^{l}, a_{m1}^{m}, a_{m1}^{u}\right) & \left(a_{m2}^{l}, a_{m2}^{m}, a_{m2}^{u}\right) & \cdots & \left(a_{mn}^{l}, a_{mn}^{m}, a_{mn}^{u}\right) \end{pmatrix} \tag{5}$$

In this matrix, $\tilde{a}_{mn}$ indicates the importance of element in the row m^{th} compared to element in the column n^{th} (Ramik, 2007). If $\tilde{A}$ is a paired-comparison matrix, it is estimated that the elements of this matrix will be reversed towards the original diameter; namely, the value of $1/a_{mn}$ can be allocated to the element of $\tilde{a}_{mn}$ (Arif and Alam, 2014; Tung and Tang, 1998). Therefore, paired comparison matrix will be the same as equation (6):

$$\tilde{A} = \begin{pmatrix} (1,1,1) & \left(a_{12}^{l}, a_{12}^{m}, a_{12}^{u}\right) & \cdots & \left(a_{1n}^{l}, a_{1n}^{m}, a_{1n}^{u}\right) \\ \left(\frac{1}{a_{21}^{u}}, \frac{1}{a_{21}^{m}}, \frac{1}{a_{21}^{l}}\right) & (1,1,1) & \cdots & \left(a_{2n}^{l}, a_{2n}^{m}, a_{2n}^{u}\right) \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{1}{a_{m1}^{u}}, \frac{1}{a_{m1}^{m}}, \frac{1}{a_{m1}^{l}}\right) & \left(\frac{1}{a_{m2}^{u}}, \frac{1}{a_{m2}^{m}}, \frac{1}{a_{m2}^{l}}\right) & \cdots & (1,1,1) \end{pmatrix} \tag{6}$$

In this method, fuzzy geometric mean is used to conclude experts' views that expressed in the format of matrices $\tilde{A}$ and paired-comparisons. There are many methods to estimate fuzzy weights of $\tilde{w}_i$ based on matrix $\tilde{A}$ with approximate value of $\tilde{a}_{ij} \approx \tilde{w}_i / \tilde{w}_j$ as $\tilde{w}_i = (w_i^l, w_i^m, w_i^u)$, i = 1,2,…,n (Kutlu and Ekmekçioğlu, 2012; Chen et al., 1992). One of these methods is logarithm least squares method based on fuzzy weights calculation in this study. In this method, triangular fuzzy weights can be calculated for criteria, sub-criteria and options such that weights output of this method can be used in fuzzy approach to rank options (Önüt et al., 2009). To obtain the weight of the above criteria due to super matrix w that can be as equation (7), the following stages were performed to calculate w_{21}, w_{21} and the corresponding weight.

$$w = \begin{pmatrix} 0 & 0 \\ w_{21} & w_{22} \end{pmatrix} \tag{7}$$

In matrix w and w_{21}, geometric mean matrix of decision maker's views is about paired-comparison of ranking indicators towards the main objective. w_{22} is also calculated from geometric mean of decision makers' views on paired-comparison of indicators towards each of other indicators (control indicator) and by use of logarithm least squares method to integrate n tables of n criteria (Önüt et al., 2009). Fuzzy weights logarithmic least squares method is shown as equation (8):

$$\tilde{w}_k = \left(w_k^l, w_k^m, w_k^u\right), k = 1,2,\ldots,n. \tag{8}$$

So that:

$$w_k^s = \frac{\left(\prod_{j=1}^{n} a_{ij}^{s}\right)^{1/n}}{\sum_{k=1}^{n}\left(\prod_{j=1}^{n} a_{kj}^{m}\right)^{1/n}}, \quad s \in \{l, m, n\} \tag{9}$$

Then, matrix w_i is calculated as $w_i = w_{22} \times w_{21}$ and fuzzy weight of each of priority indictors obtained using logarithm least squares method.

3.2 Fuzzy TOPSIS method: fuzzy techniques for order preference by similarity to idea solution

Fuzzy TOPSIS is widely used as one of MADM fuzzy techniques, due to its simplicity and programmable nature (Kutlu and Ekmekçioğlu, 2012). The basis of this technique is to determine the best option based on the concept of ideal solution. Thus, the option with the maximum distance from negative ideal and the minimum distance from positive ideal is in a higher priority (Yuen and Fung, 2014; Kannan et al., 2014). Developed method (Chen, 2000) was used in this study; it involves the following seven steps:

Step 1 Formation of fuzzy decision making matrix, according to equation (10):

$$\tilde{D} = \begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_m \\ \tilde{w} \end{array} \begin{array}{c} \begin{array}{cccc} C_1 & C_2 & \cdots & C_n \end{array} \\ \begin{pmatrix} \tilde{x}_{11} & \tilde{x}_{12} & \cdots & \tilde{x}_{1n} \\ \tilde{x}_{21} & \tilde{x}_{22} & \cdots & \tilde{x}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{x}_{m1} & \tilde{x}_{m2} & \cdots & \tilde{x}_{mn} \\ \tilde{w}_1 & \tilde{w}_2 & \cdots & \tilde{w}_n \end{pmatrix} \end{array} \tag{10}$$

In this matrix, $A = \{A_i | i = 1,2,\ldots,m\}$ are the options that should be selected by decision makers among their considered options. $C = \{C_j | j = 1,2,\ldots,n\}$ is the criterion that the options are measured by it.

$\tilde{X} = \{\tilde{x}_{ij} | i = 1,2,\ldots,m;\ j = 1,2,\ldots,n\}$ is the viewpoint of the ith people on the jth component that is calculated as fuzzy number of equation (11):

$$\tilde{x}_{ij} = \left(a_{ij}, b_{ij}, c_{ij}\right) \tag{11}$$

And finally, $\tilde{w} = \{\tilde{w}_j \mid j = 1,2,\ldots,n\}$ that includes the set of fuzzy weight of criteria is required as an input for fuzzy TOPSIS method. It should be noted that the weight of each criterion in this research was obtained using FANP method.

Step 2 Defuzzification of decision making matrix.

At this step, fuzzy decision making matrix for evaluation of options should be converted into a normalised fuzzy matrix $\tilde{R}$.

Equation (12) is used to obtain this matrix, where m is the number of options and n the number of experts.

$$\tilde{R} = \left[\tilde{r}_{ij}\right]_{m \times n} \quad i = 1,2,\ldots,m;\ j = 1,2,\ldots,n. \tag{12}$$

If fuzzy numbers are as (a, b, c) defuzzification matrix (normalised) is obtained through equation (13):

$$\tilde{r}_{ij} = \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*}\right) \tag{13}$$

In equation (9), c_j^* is the maximum value of c, j^{th} expert among all options. Equation (13) indicates this issue.

$$c_j^* - \max_i c_{ij} \tag{14}$$

Step 3 Calculation of the weighted normalised fuzzy matrix $\tilde{V}$: at this stage, the normalised matrix of paired-comparisons is weighted by equation (15).

$$\tilde{V} = \left[\tilde{v}_{ij}\right]_{m\times n}; \quad i = 1,2,\ldots,m;\ j = 1,2,\ldots,n. \tag{15}$$

$$\tilde{v}_{ij} = \tilde{r}_{ij} \otimes \tilde{w}_j \tag{16}$$

In this equation, $\tilde{r}_{ij}$ is the normalised matrix obtained from second step.

Step 4 Determination of fuzzy positive ideal solution (*FPIS*, A^+) and fuzzy negative ideal (*FPIS*, A^-) solution: the basis of FTOPSIS method is to calculate the distance level of options from fuzzy positive and negative ideal solution. Therefore at this stage, fuzzy positive and negative ideal solutions are determined. Equations (17) and (18) are used to determine these solutions.

$$\begin{aligned} FPIS = \tilde{A}^+ &= \left\{\tilde{v}_1^+(x), \tilde{v}_2^+(x),\ldots,\tilde{v}_j^+(x),\ldots\tilde{v}_m^+(x)\right\} \\ &= \left\{\left(\max_i \tilde{v}_{ij}(x)\middle| j \in J_1\right), \left(\min_i \tilde{v}_{ij}(x)\middle| j \in J_2\right)\middle| i = 1,2,\ldots,n\right\} \end{aligned} \tag{17}$$

$$\begin{aligned} FNIS = \tilde{A}^- &= \left\{\tilde{v}_1^-(x), \tilde{v}_2^-(x),\ldots,\tilde{v}_j^-(x),\ldots\tilde{v}_m^-(x)\right\} \\ &= \left\{\left(\min_i \tilde{v}_{ij}(x)\middle| j \in J_1\right), \left(\max_i \tilde{v}_{ij}(x)\middle| j \in J_2\right)\middle| i = 1,2,\ldots,n\right\} \end{aligned} \tag{18}$$

where j_1 and j_2 are related to positive and negative criteria, respectively. It should be noted that fuzzy positive and negative ideal values introduced by Chen are used in this study. These values are:

$$v_j^+ = (1, 1, 1) \tag{19}$$

$$v_j^- = (0, 0, 0) \tag{20}$$

Step 5 Calculating the distance level of each of options from fuzzy positive and negative ideal solutions.

If $\tilde{A}$ and $\tilde{B}$ are two fuzzy numbers as follows; then, the distance between these two fuzzy numbers is obtained by equation (21):

$$\tilde{A} = \left(a_1, b_1, c_1\right)$$

$$\tilde{B} = \left(a_2, b_2, c_2\right)$$

$$D(\tilde{A}, \tilde{B}) = \sqrt{\frac{1}{3}\left[\left(a_2 - a_1\right)^2 + \left(b_2 - b_1\right)^2 + \left(c_2 - c_1\right)^2\right]} \tag{21}$$

According to the above descriptions on the way of calculating the distance between two fuzzy numbers, the distance of each of factors from positive and negative ideals is obtained by equations (22) and (23):

$$d_i^+ = \sum_{j=1}^{n} d\left(\tilde{v}_{ij} - v_{ij}^+\right); \quad i = 1, 2, \ldots, m. \tag{22}$$

$$d_i^- = \sum_{j=1}^{n} d\left(\tilde{v}_{ij} - v_{ij}^-\right); \quad i = 1, 2, \ldots, m. \tag{23}$$

Step 6 Calculating relative proximity of ith option from ideal solution.

This relative proximity is defined as equation (24):

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-}; \quad i = 1, 2, \ldots, n. \tag{24}$$

Step 7 Ranking of options.

The options existing in the issue can be ranked based on descending order of CC_i. Each option which has a larger CC_i, is a priority.

4 Research results

After collecting data of expert survey questionnaire; according to the conducted calculation, the weight of each of the criteria was obtained by use of fuzzy ANP method. The results are shown in the following table.

Table 3 The weight of criteria affecting the improvement of hotel industry service quality

Dimensions	*Final fuzzy weight*	*Final decisive weight of dimensions*
Physical and tangible factors	(0.246, 0.314, 0.365)	0.311
Creating trust and confidence	(0.127, 0.189, 0.224)	0.184
Accountability	(0.118, 0.151, 0.2)	0.154
Guaranteed services	(0.133, 0.177, 0.225)	0.178
Empathy and observing Islamic values	(0.129, 0.168, 0.211)	0.169

As it is observed in Table 3, among different criteria of model, physical and tangible factors with the weight of 0.311, creating trust and confidence with the weight of 0.186 and guaranteed services with the weight of 0.178 were recognised as having the highest weight. In continue, by inserting the weights obtained from Table 3 in fuzzy TOPSIS method; prioritisation of the factors affecting the improvement of hotel industry service quality is achieved according to Table 4.

Table 4 Prioritisation of the factors affecting the improvement of hotel industry service quality

NO	*Factors*	di^+	di^-	*CC*	*Rank*
1	Modernity of hotel facilities	159.292	0.723	0.00452	1
2	Attractiveness of physical facilities in the hotel	159.337	0.677	0.00423	20
3	Dress up of staffs	159.299	0.710	0.00444	9
4	Locating hotel in a good climate region	159.333	0.683	0.00427	18
5	A balance between number of staffs and their responsibilities in the hotels	159.310	0.703	0.00439	13
6	The cleanliness of hotel environment	159.320	0.696	0.00435	14
7	Full performance of obligations	159.307	0.709	0.00443	10
8	Proper delivery of service by staffs	159.293	0.722	0.00451	6
9	Delivery of high quality service at the time promised	159.297	0.719	0.00449	7
10	Security of passengers	159.301	0.713	0.00446	8
11	Staffs' tendency in providing fast and reliable services to customers	159.321	0.695	0.00434	15
12	Staffs' tendency in solving customers' problems	159.308	0.708	0.00442	11
13	Appropriate response time	159.332	0.685	0.00428	17
14	Staffs' focus on addressing the needs and demands of customers	159.332	0.680	0.00425	19
15	The ability and credibility of hotel in providing reliable and good services	159.284	0.729	0.00456	4
16	Having the qualified and skilled staffs for hotel service delivery	159.292	0.726	0.00454	2
17	Customers' trust in the hotel staffs	159.284	0.731	0.00457	5
18	Staffs' respect to customers' character	159.280	0.735	0.00459	3
19	Particular attention of staffs to cultural differences of customers	159.310	0.705	0.00441	12
20	Understanding customers' needs by hotel staffs	159.329	0.688	0.00430	16
21	Politeness and humility of the hotel staffs	159.332	0.655	0.00409	21

As it is observed in Table 4, among the factors of the improvement of service quality; Modernity of hotel facilities, Having the qualified and skilled staffs for hotel service delivery and Staffs' respect to customers' character were respectively identified as the most important factors affecting the improvement of hotel industry service quality; and on the other side, the factors Staffs' focus on addressing the needs and demands of customers, Attractiveness of physical facilities in the hotel and Politeness and humility of the hotel staffs were placed at the lowest level of ranking.

5 Conclusions

In fact, quality is what the customer wants; in other words, a high quality product should be in compliance with customers' needs (Prybutok, 2012). Service quality is a background for customer's satisfaction. In today's competitive markets, reputation and profitability of a hotel has a direct and strong correlation with level of customer's satisfaction and customers' satisfaction leads to increase loyalty and therefore maintaining the customer relationship (Stefano et al., 2015). Level of customers' satisfaction; in addition to revealing the degree of success in achieving the goals of a hotel, improve the quality of services in the hotel. One of the most important actions in attracting customer's satisfaction is to measure his/her level of satisfaction from receiving services; this level can be measured with implementation of appropriate measurement programs. The goal of conducting this study is to identify and rank the factors affecting the improvement of the quality of hotel industry services. Very few studies have been conducted in identifying and prioritising the factors affecting the improvement of the quality of hotel industry services, besides deterministic MADM methods have been commonly used in ranking the elements of the quality of services. In this work, considering the advantages of fuzzy logic in correctly reflecting the human views, we use fuzzy ANP and fuzzy TOPSIS methods, as an innovative way of ranking of the quality of services. According to the results of this study, modernising and updating of hotel facilities, having professional staffs for delivery of hoteling services and staffs' respect to customers' character were identified as the most important factors affecting the improvement of hotel industry services. Due to the facilities of four and five-star hotels of Mashhad city, most foreign passengers who were staying at these hotels complained about poor and inefficient equipment of them. They believed that there are not professional staffs at these hotels and the majority of them were dissatisfied about the delivered service. Therefore, it is suggested that managers of these hotel pay more attention to the above factors. It is recommended to these managers that get more satisfaction of their customers, enjoying from the views of successful domestic and foreign consultants in the field of hotel service delivery. Because, if the level of passengers' satisfaction decreases; waiting is far away to return them again to these hotels in far-coming years. Every study has its own limitation and this study is not an exception. The followings are of the limitations of this study:

- lack of scientific resources in this field
- problems in convincing the people for filling the questionnaire
- lack of some managers' cooperation due to not believing in these types of researches.

The followings are suggested for further studies:

- It is suggested to do similar researches for other vital industries such as banking industry, insurance, etc. Because defining the best factors affecting the improvement of the quality services can increase the attraction of customers in the market.
- Considering the fact that the combination of fuzzy ANP and fuzzy TOPSIS are used in this study, it is suggested that the fuzzy AHP will be used in further researches to analyse the dependence of the elements.

References

Ali, F. and Amin, M. (2014) 'The influence of physical environment on emotions, customer satisfaction and behavioural intentions in Chinese resort hotel industry', *Journal for Global Business Advancement*, Vol. 7, No. 3, pp.249–266.

Alizadeh, K. (2011) 'The role of historic-religious monuments in the future economy of Mashhad City', *The Annual conference of Canadian Association of Geographers*, Calgary, Alberta, Canada, May.

Al-Refaie, A. (2015) 'Effects of human resource management on hotel performance using structural equation modeling', *Computers in Human Behavior*, February, Vol. 43, No. C, pp.293–303.

Amritharaj, S.M.D. and Vembar, V. (2014) 'Demographic and environment factors influence on training and development effectiveness in hotel industry: a case study of selected hotels in Chennai', *Journal of Human Resources*, Vol. 2, No. 1, pp.83–95.

Anca, M. and Andreea, N.N. (2013) 'Service quality analysis-Hotel Alpin (POIANA BRASOV)', *Analele Universităţii Constantin Brâncuşi din Târgu Jiu: Seria Economie*, Vol. 6, No. 6, pp.267–270, Academica Brâncuşi, ISSN: 1844-7007 (Print).

Aniley, B. and Negi, R. (2015) 'Service quality attributes affecting passengers' satisfaction with HIGER city bus', *Ethiopian Journal of Business and Economics*, Vol. 1, No. 2, pp.70–98.

Arif, R.B. and Alam, M.N. (2014) 'Ranking of alternative information sources using AHP model in a developing setting: evidence from individual investors of the Dhaka Stock Exchange (DSE) of Bangladesh', *Journal of Business Management & Social Sciences Research*, Vol. 3, No. 12, pp.64–69.

Ariffin, A.A.M. and Maghzi, A. (2012) 'A preliminary study on customer expectations of hotel hospitality: influences of personal and hotel factors', *International Journal of Hospitality Management*, Vol. 31, No. 1, pp.191–198.

Attallah, N.F. (2015) 'Evaluation of perceived service quality provided by tourism establishments in Egypt', *Tourism and Hospitality Research*, Vol. 15, No. 3, pp.149–160.

Aznar, J., Ferrís-Oñate, J. and Guijarro, F. (2010) 'An ANP framework for property pricing combining quantitative and qualitative attributes', *Journal of the Operational Research Society*, Vol. 61, No. 5, pp.740–755.

Babakus, E. and Inhofe, M. (2015) 'Measuring perceived service quality as a multi-attribute attitude', Paper read at *Proceedings of the 1993 Academy of Marketing Science (AMS) Annual Conference*.

Bansal, H.S. and Taylor, S. (2015a) 'Investigating the relationship between service quality, satisfaction and switching intentions', Paper read at *Proceedings of the 1997 Academy of Marketing Science (AMS) Annual Conference.*

Bansal, H.S. and Taylor, S.F. (2015b) 'Beyond service quality and customer satisfaction: investigating additional antecedents of service provider switching intentions', Paper read at *Proceedings of the 1999 Academy of Marketing Science (AMS) Annual Conference*.

Betsch, T. and Haberstroh, S. (2014) *The Routines of Decision Making*, Psychology Press, Mahwah, New Jersey.

Blešić, I., Čerović, S. and Dragićević, V. (2011) 'Improving the service quality as a socially responsible activity of hotel companies', *Amfiteatru Economic*, Vol. 13, No. 29, pp.273–286.

Bojanic, D.C. (2015) 'An examination of the means-end model of price, quality and value in the service industry', Paper read at *Proceedings of the 1996 Academy of Marketing Science (AMS) Annual Conference*.

Borkar, S. and Koranne, S. (2014) 'Study of service quality management in hotel industry', *Pacific Business Review International*, Vol. 6, No. 9, pp.21–25.

Caruana, A. and Ramaseshan, B. (2015a) 'The effect of service quality and consumer trust on retail website loyalty', in *Creating and Delivering Value in Marketing*, pp.76–76, Springer, Cham, DOI: 10.1007/978-3-319-11848-2_21.

Caruana, A., Pitt, L.F. and Ramasashan, B. (2015b) 'Satisfaction as a mediator in the link between service quality and service loyalty', Paper read at *Proceedings of the 2000 Academy of Marketing Science (AMS) Annual Conference*.

Chadha, A. (2015) 'Case study of hotel Taj in the context of CRM and customer retention', *Kuwait Chapter of the Arabian Journal of Business and Management Review*, Vol. 4, No. 7, p.1.

Chang, B., Kuo, C., Wu, C-H. and Tzeng, G-H. (2015a) 'Using fuzzy analytic network process to assess the risks in enterprise resource planning system implementation', *Applied Soft Computing*, Vol. 28, No. C, pp.196–207.

Chang, Y-H., Shao, P.C. and Chen, H.J. (2015b) 'Performance evaluation of airport safety management systems in Taiwan', *Safety Science*, 30 June, Vol. 75, pp.72–86.

Chang, K.C., Chen, M.C., Kuo, N.T., Hsu, C.L. and Cheng, Y.S. (2016) 'Applying data mining methods to tourist loyalty intentions in the international tourist hotel sector', *Anatolia*, Vol. 27, No. 2, pp.271–274.

Chen, C-M., Lin, Y-C. and Liao, Y-H. (2014) 'Research note: revenue and labour demand in the hotel industry: an investigation of endogeneity', *Tourism Economics*, Vol. 20, No. 2, pp.439–446.

Chen, C-M., Yeh, C-Y. and Hu, J.L. (2011) 'Influence of uncertain demand on product variety: evidence from the international tourist hotel industry in Taiwan', *Tourism Economics*, Vol. 17, No. 6, pp.1275–1285.

Chen, C-T. (2000) 'Extensions of the TOPSIS for group decision-making under fuzzy environment', *Fuzzy sets and systems*, Vol. 114, No. 1, pp.1–9.

Chen, I-S. (2012) 'A revised Inno-Qual performance system for higher education: the integrated applications of DEMATEL and ANP', *Journal of the Operational Research Society*, Vol. 63, No. 4, pp.478–488.

Chen, S.J., Hwang, C.L. and Hwang, F.P. (1992) 'Fuzzy multiple attribute decision making, *Lecture Notes in Economics and Mathematical Systems*, vol. 375.

Chen, Y-T. and Chang, C.T. (2014) 'Multi-coefficient goal programming in thinning schedules to increase carbon sequestration and improve forest structure', *Annals of Forest Science*, Vol. 71, No. 8, pp.907–915.

Cronin Jr., Joseph, J. and Taylor, S.A. (1992) 'Measuring service quality: a reexamination and extension', *The Journal of Marketing*, Vol. 56, No. 3, pp.55–68.

Cronin, J.J., Brady, M.K. and Hult, G.T.M. (2000) 'Assessing the effects of quality, value, and customer satisfaction on consumer behavioral intentions in service environments', *Journal of Retailing*, Vol. 76, No. 2, pp.193–218.

Dabholkar, P.A. (2015) 'How to improve perceived service quality by increasing customer participation', *Paper read at Proceedings of the 1990 Academy of Marketing Science (AMS) Annual Conference*.

Dedeoglu, B.B.H. et al. (2015) 'Differences in service quality perceptions of stakeholders in the hotel industry', *International Journal of Contemporary Hospitality Management*, Vol. 27, No. 1, pp.130–146

Deng, C. and Zuo, L. (2014) 'A comparative study on hotel services marketing differentiation between Beijing and Taipei', Paper read at *11th International Conference on Service Systems and Service Management (ICSSSM)*.

Dhar, R.L. (2015) 'Service quality and the training of employees: the mediating role of organizational commitment', *Tourism Management*, 247 August, Vol. 46, pp.419–430.

Dortyol, T., Ibrahim, I.V. and Kitapci, O. (2014) 'How do international tourists perceive hotel quality? An exploratory study of service quality in Antalya tourism region', *International Journal of Contemporary Hospitality Management*, Vol. 26, No. 3, pp.470–495.

Dubé, L. (2015) 'An experimental investigation of the effect of perceived service quality on consumer satisfaction with service encounters', Paper read at *Proceedings of the 1992 Academy of Marketing Science (AMS) Annual Conference*.

Gogus, O. and Boucher, T.O. (1998) 'Strong transitivity, rationality and weak monotonicity in fuzzy pairwise comparisons', *Fuzzy Sets and Systems*, Vol. 94, No. 1, pp.133–144.

Gottwald, S. (2015) 'The logic of fuzzy set theory: a historical approach', in *Petr Hájek on Mathematical Fuzzy Logic*, pp.41–61, Springer, Springer International Publishing Switzerland, DOI: 10.1007/978-3-319-06233-4_3, Online ISBN: 978-3-319-06233-4.

Gremler, D.D. (2011) 'SMA distinguished teacher comment teaching (services) marketing: walking the talk', *Advances in Marketing*, Vol. 25, No. 1, p.8.

Guesalaga, R. and Pitta, D. (2014) 'The importance and formalization of service quality dimensions: a comparison of Chile and the USA', *Journal of Consumer Marketing*, Vol. 31, No. 2, pp.145–151.

Gummesson, E., Mele, C., Polese, F., Gummesson, E. and Grönroos, C. (2012) 'The emergence of the new service marketing: Nordic School perspectives', *Journal of Service Management*, Vol. 23, No. 4, pp.479–497.

Gupta, N. and Raman, R. (2014) 'Selection of Microstrip patch antenna substrate for WLAN application using multiple attribute decision making approach', *International Journal of Microwave Science and Technology*, October, p.5.

Haksever, C., Cook, R.G. and Chaganti, R. (2015) 'Applicability of the gaps model to service quality in small firms', *Journal of Small Business Strategy*, Vol. 8, No. 1, pp.49–66.

Hanzaee, K. and Mirvaisi, M. (2013) 'A survey on impact of emotional intelligence, organizational citizenship behaviors and job satisfaction on employees' performance in Iranian hotel industry', *Management Science Letters*, Vol. 3, No. 5, pp.1395–1402.

Hapsari, R., Clemes, M. and Dean, D. (2016) 'The mediating role of perceived value on the relationship between service quality and customer satisfaction: evidence from indonesian airline passengers', *Procedia Economics and Finance*, Vol. 35, pp.388–395.

Hopkins, C.D., Mitra, K. and Engelland, B.T. (2015). 'Effects of salesperson attractiveness on consumer perceptions of service quality: a conceptual framework', Paper read at *Proceedings of the 1998 Academy of Marketing Science (AMS) Annual Conference*.

Huang, J-J. (2012) 'A mathematical programming model for the fuzzy analytic network process – applications of international investment', *Journal of the Operational Research Society*, Vol. 63, No. 11, pp.1534–1544.

Huei, C.T. and Easvaralingam, Y. (2011) 'Perceptions of service quality, corporate image, and customer loyality in the hotel industry of Malaysia', Paper read at *2nd International Research Symposium in Service Management*, Yogyakarta, Indonesia.

Izogo, E.E. and Ogba, I-E. (2015) 'Service quality, customer satisfaction and loyalty in automobile repair services sector', *International Journal of Quality & Reliability Management*, Vol. 32, No. 3, pp.250–269.

Jayawardhena, C. (2015) 'Customers' service evaluation: antecedent effects of employees' customer orientation and service orientation behaviors', Paper read at *Proceedings of the 2009 Academy of Marketing Science (AMS) Annual Conference*.

Kannan, D., Lopes de Sousa Jabbour, A.B. and Jabbour, C.J.C. (2014) 'Selecting green suppliers based on GSCM practices: using fuzzy TOPSIS applied to a Brazilian electronics company', *European Journal of Operational Research*, Vol. 233, No. 2, pp.432–447.

Karjaluoto, H., Jayawardhena, C., Pihlström, M. and Leppäniemi, M. (2015) 'Effects of service quality, trust, and perceived value on customer loyalty: the case of mobile subscribers', *Proceedings of the 2009 Academy of Marketing Science (AMS) Annual Conference*, Springer International Publishing, January, pp.179–179.

Karwowski, W. and Mital, A. (2014) *Applications of Fuzzy Set Theory in Human Factors*, Elsevier, University of Michigan.

Kilic, H.S., Zaim, S. and Delen, D. (2015) 'Selecting – the best: ERP system for SMEs using a combination of ANP and PROMETHEE methods', *Expert Systems with Applications*, Vol. 42, No. 5, pp.2343–2352.

Kozak, M.A. and Gürel, D.A. (2015) 'Service design in hotels: a conceptual review', *Tourism*, Vol. 63, No. 2, pp225–240.

Kutlu, A.C. and Ekmekçioğlu, M. (2012) 'Fuzzy failure modes and effects analysis by using fuzzy TOPSIS-based fuzzy AHP', *Expert Systems with Applications*, Vol. 39, No. 1, pp.61–67.

Li, G., Kou, G., Lin, C., Xu, L. and Liao, Y. (2015a) 'Multi-attribute decision making with generalized fuzzy numbers', *Journal of the Operational Research Society*, Vol. 66, No. 11, pp.1793–1803.

Li, G., Law, R., Vu, H.Q., Rong, J. and Zhao, X.R. (2015b) 'Identifying emerging hotel preferences using emerging pattern mining technique', *Tourism Management*, June, Vol. 46, pp.311–321, DOI: 10.1016/j.tourman.2014.06.015.

Liat, C.B., Mansori, S. and Huei, C.T. (2014) 'The associations between service quality, corporate image, customer satisfaction, and loyalty: evidence from the Malaysian hotel industry', *Journal of Hospitality Marketing & Management*, Vol. 23, No. 3, pp.314–326.

Lin, C-F. (2014) 'Exploring the hotel service personnel's cognitive implications toward service attributes and ethics', *Human Factors and Ergonomics in Manufacturing & Service Industries*, Vol. 24, No. 1, pp.14–28.

Lotfi, F.H. and Fallahnejad, R. (2010) 'Imprecise Shannon's entropy and multi attribute decision making', *Entropy*, Vol. 12, No. 1, pp.53–62.

Lun, Y.V., Shang, K.C., Lai, K.H. and Cheng, T.C.E. (2016) 'Examining the influence of organizational capability in innovative business operations and the mediation of profitability on customer satisfaction: an application in intermodal transport operators in Taiwan', *International Journal of Production Economics*, Vol. 171, No. 2, pp.179–188.

Maier, K., Frysak, J. and Bernroider, E.W. (2014) 'The effects of effort-feedback on time spent for information processing in multi-criteria decision making', in *New Perspectives in Information Systems and Technologies*, Vol. 1, pp.535–544, Springer International Publishing, Springer, Cham, DOI: 10.1007/978-3-319-05951-8_51.

Markovic, S., Raspor, S., Ivankovic, G. and Planinc, T. (2013) 'A study of expected and perceived service quality in Croatian and Slovenian hotel industry', *European Journal of Tourism Research*, Vol. 6, No. 1, p36.

Martínez, P. and del Bosque, I.R. (2013) 'CSR and customer loyalty: the roles of trust, customer identification with the company and satisfaction', *International Journal of Hospitality Management*, December, Vol. 35, pp.89–99.

Minh, N.H., Ha, N.T., Chi Anh, P. and Matsui, Y. (2015) 'Service quality and customer satisfaction: a case study of hotel industry in Vietnam', *Asian Social Science*, Vol. 11, No. 10, p.73.

Mohsin, A. and Lengler, J. (2015) 'Service experience through the eyes of budget hotel guests: do factors of importance influence performance dimensions?', *Journal of Hospitality and Tourism Management*, June, Vol. 23, pp.23–34.

Molina-Azorín, J.F., Tarí, J.J., Pereira-Moliner, J., López-Gamero, M.D. and Pertusa-Ortega, E.M. (2015) 'The effects of quality and environmental management on competitive advantage: a mixed methods study in the hotel industry', *Tourism Management*, October, Vol. 50, pp.41–54.

Mueller, H. and Kaufmann, E.L. (2001) 'Wellness tourism: market analysis of a special health tourism segment and implications for the hotel industry', *Journal of Vacation Marketing*, Vol. 7, No. 1, pp.5–17.

Nasution, H. (2016) 'Coaligning service quality attributes and its implication to customer value', in *Looking Forward, Looking Back: Drawing on the Past to Shape the Future of Marketing*, pp.751–759, Springer International Publishing, Monash University, Australia.

Nejati, M., Nejati, m. and Shafaei, a. (2009) 'Ranking airlines' service quality factors using a fuzzy approach: study of the Iranian society', *International Journal of Quality & Reliability Management*, Vol. 26, No. 3, pp.247–260.

Önüt, S., Kara, S.S. and Işik, E. (2009) 'Long term supplier selection using a combined fuzzy MCDM approach: a case study for a telecommunication company', *Expert Systems with Applications*, Vol. 36, No. 2, pp.3887–3895.

Parasuraman, A., Zeithaml, V.A. and Berry, L.L. (1985) 'A conceptual model of service quality and its implications for future research', *The Journal of Marketing*, Vol. 49, No. 4, pp.41–50.

Pastor-Ferrando, J.P., Aragonés-Beltrán, P., Hospitaler-Pérez, A. and García-Melón, M. (2010) 'An ANP-and AHP-based approach for weighting criteria in public works bidding', *Journal of the Operational Research Society*, Vol. 61, No. 6, pp.905–916.

Patil, S.K. and Kant, R. (2014) 'A hybrid approach based on fuzzy DEMATEL and FMCDM to predict success of knowledge management adoption in supply chain', *Applied Soft Computing*, May, Vol. 18, pp.126–135.

Paul, J., Mittal, A. and Srivastav, G. (2016) 'Impact of service quality on customer satisfaction in private and public sector banks', *International Journal of Bank Marketing*, Vol. 34, No. 5, pp.606–622.

Peng, K-H. and Tzeng, G-H. (2013) 'A hybrid dynamic MADM model for problem-improvement in economics and business', *Technological and Economic Development of Economy*, Vol. 19, No. 4, pp.638–660.

Pijls, R. and Groen, B.H. (2012) 'Cleanliness translated into sensory clues of the service environment', Paper read at *Proceedings of the 11th EuroFM Research Symposium*.

Prakash, N., Somasundaram, R. and Krishnamoorthy, V. (2016) 'A study on linkage between service quality and customer satisfaction in hotel industry', *Asian Journal of Research in Social Sciences and Humanities*, Vol. 6, No. 4, pp.285–295.

Prince, P.R. and Khaleq, Z.B. (2013) 'Assessment of gap between service quality expectation and perception: a study on the walk-in guests of economic Hotels in Cox's Bazar Bangladesh', *The IUP Journal of Marketing Management*, Vol. 12, No. 3, pp.8–26.

Prybutok, V.R. (2012) *The Impact of Quality on Customer Behavioral Intentions Based on the Consumer Decision Making Process as Applied in E-Commerce*, Doctoral Dissertation, University of North Texas, ISBN: 978-1-267-99964-1.

Rahimi, R. and Kozak, M. (2016) Impact of customer relationship management on customer satisfaction: the case of a budget hotel chain', *Journal of Travel & Tourism Marketing*, Vol. 34, No. 1, pp.1–12.

Ramik, J. (2007) 'A decision system using ANP and fuzzy inputs', *International Journal of Innovative Computing, Information and Control*, Vol. 3, No. 4, pp.825–837.

Rao, S. (2014) 'Dimensions of service quality in restaurants: a comparative study in Udupi and Dakshina Kannada districts', *Global Conference on Service Management*, Auro University, Surat, pp.44–46, ISBN 13: 978-81-928189-2-4.

Rauch, D.A., Collins, M.D., Nale, R.D., Barr, P.B., Okumus, F. and Okumus, F. (2015) 'Measuring service quality in mid-scale hotels', *International Journal of Contemporary Hospitality Management*, 30 September, Vol. 50, pp.576–587.

Reichel, A., Fuchs, G., Pizam, A. and Poria, Y. (2014) 'Occupational self-perceptions of hotel employees: an exploratory study', *Tourism Analysis*, Vol. 19, No. 5, pp.637–641.

Ren, L. et al. (2015) 'Understanding customer satisfaction with budget hotels through online comments: evidence from home inns in China', *Journal of Quality Assurance in Hospitality & Tourism*, Vol. 16, No. 1, pp.45–62.

Rhee, H.T. and Yang, S.-B. (2015) 'Does hotel attribute importance differ by hotel? Focusing on hotel star-classifications and customers' overall ratings', *Computers in Human Behavior*, September, Vol. 50, No. C, pp.576–587.

Saaty, T.L. and Takizawa, M. (1986) 'Dependence and independence: from linear hierarchies to nonlinear networks', *European Journal of Operational Research*, Vol. 26, No. 2, pp.229–237, p.10.

Salah, A. and Moselhi, O. (2015) 'Contingency modelling for construction projects using fuzzy-set theory', *Engineering, Construction and Architectural Management*, Vol. 22, No. 2, pp.214–241.

Sattar, S., Ali, I. and Arshad, A. (2014) 'Customer loyalty toward Nokia Brand', *Journal of Marketing and Consumer Research*, Vol. 4, pp.22–28.

Seçme, N.Y., Bayrakdaroğlu, A. and Kahraman, C. (2009) 'Fuzzy performance evaluation in Turkish banking sector using analytic hierarchy process and TOPSIS', *Expert Systems with Applications*, Vol. 36, No. 9, pp.11699–11709.

Sevkli, M., Oztekin, A., Uysal, O., Torlak, G., Turkyilmaz, A. and Delen, D. (2012) 'Development of a fuzzy ANP based SWOT analysis for the airline industry in Turkey', *Expert Systems with Applications*, Vol. 39, No. 1, pp.14–24.

Siavoshani, E., Dehghanan, H. and Bakhshandeh, G. (2014) 'A study on the effects of service quality on brand personality: evidence from hotel industry', *Management Science Letters*, Vol. 4, No. 7, pp.1547–1550.

Siddiqi, K.O. (2011) 'Interrelations between service quality attributes, customer satisfaction and customer loyalty in the retail banking sector in Bangladesh', *International Journal of Business and Management*, Vol. 6, No. 3, p.12.

Sitawati, R., Winata, L. and Mia, L. (2015) 'Competitive strategy and sustainable performance: the application of sustainable balanced scorecard', *Issues in Social & Environmental Accounting*, Vol. 9, No. 1, pp51–75.

Sofi, M.R., Hakim, I.A. and Gadoo, M.R. (2016) 'Service quality variables and tourist satisfaction at destination level – a study of J&K tourism', *International Journal of Applied Research and Studies*, Vol. 3, No. 2, pp25–30.

Sohrabi, B., Vanani, I.R., Tahmasebipur, K. and Fazli, S. (2012) 'An exploratory analysis of hotel selection factors: a comprehensive survey of Tehran hotels', *International Journal of Hospitality Management*, Vol. 31, No. 1, pp.96–106.

Sorayaei, A., Ruhollahi, M., Ashrafi, S., Khor, B.H. and RezaZadeh, A. (2014) 'Using fuzzy AHP to determinate the marketing strategy for the private hotels in Iran', *Advances in Environmental Biology*, Vol. 8, No. 2, pp.360–368.

Stefano, N.M., Filho, N.C., Barichello, R. and Sohn, A.P. (2015) 'A fuzzy SERVQUAL based method for evaluated of service quality in the hotel industry', *Procedia CIRP*, Vol. 30, pp.433–438.

Tadić, S., Zečević, S. and Krstić, M. (2014) 'A novel hybrid MCDM model based on fuzzy DEMATEL, fuzzy ANP and fuzzy VIKOR for city logistics concept selection', *Expert Systems with Applications*, Vol. 41, No. 18, pp.8112–8128.

Tavana, M. and Sodenkamp, M.A. (2010) 'A fuzzy multi-criteria decision analysis model for advanced technology assessment at Kennedy Space Center', *Journal of the Operational Research Society*, Vol. 61, No. 10, pp.1459–1470.

Tsai, Y-H., Wu, C-T. and Chen, S-L. (2015) 'Hotel choice criteria by business and leisure travelers', *The International Journal of Organizational Innovation*, Vol. 7, No. 4, pp158–164.

Tung, S.L. and Tang, S.L. (1998) 'A comparison of the Saaty's AHP and modified AHP for right and left eigenvector inconsistency', *European Journal of Operational Research*, Vol. 106, No. 1, pp.123–128.

Tzeng, G-H. and Huang, J-J. (2011) *Multiple Attribute Decision Making: Methods and Applications*, 352pp, 22 June, Chapman and Hall/CRC, ISBN 9781439861578-CAT# K12846.

Vala, M. (2014) *The Selected Antecedents and Outcomes of Work Engagement: A Study on Frontline Employees of 3, 4 and 5 Star Hotels in Iran (Tehran)*, Eastern Mediterranean University (EMU)-Doğu Akdeniz Üniversitesi (DAÜ).

Varesi, H.R., Tghvaei, M. and Shahivandi, A. (2012) 'An analysis of status of tourism infrastructures in Isfahan (with emphasis putting on hotels)', *Geography and Environmental Planning*, Vol. 22, No. 4, pp.91–112.

Walker, J.L. (1995) 'Service encounter satisfaction: conceptualized', *Journal of Services Marketing*, Vol. 9, No. 1, pp.5–14.

Wan, S-P. and Li, D-F. (2013) 'Fuzzy LINMAP approach to heterogeneous MADM considering comparisons of alternatives with hesitation degrees', *Omega*, Vol. 41, No. 6, pp.925–940.

Wang, C-H., Chen, K-Y. and Chen, S-C. (2012) 'Total quality management, market orientation and hotel performance: the moderating effects of external environmental factors', *International Journal of Hospitality Management*, Vol. 31, No. 1, pp.119–129.

Wang, J., Hooper, K. and Sinclair, R.M.S. (2013) *The Productive and Symbolic Functions of Internal Control in the Chinese Hotel Industry*, p.23, 29 November, The Department of Accounting and Finance, Unitec Institute of Technology.

Whedon, J.M., Punzo, M., Dehen, R., Menard, M.B., Fogel, D. and Olejownik, J. (2016) 'Relevance of quality measurement to integrative healthcare in the United States', *The Journal of Alternative and Complementary Medicine*, Vol. 22, No. 11, pp. –53-858.

Yang, C-C., Jou, Y-T. and Cheng, L-Y. (2011) 'Using integrated quality assessment for hotel service quality', *Quality & Quantity*, Vol. 45, No. 2, pp.349–364.

Yaya, P.L.H., Marimon, F. and Casadesus, M. (2014) 'Customer satisfaction and the role of demographic characteristics in online banking', in *Evaluating Websites and Web Services: Interdisciplinary Perspectives on User Satisfaction*, *International Journal for Quality Research*, pp.138–154, IGI Global.

Ye, Q., Li, H., Wang, Z. and Law, R. (2014) 'The influence of hotel price on perceived service quality and value in e-tourism an empirical investigation based on online traveler reviews', *Journal of Hospitality & Tourism Research*, Vol. 38, No. 1, pp.23–39.

Yu, M-C., Keng, I. and Chen, H-X. (2015) 'Measuring service quality via a fuzzy analytical approach', *International Journal of Fuzzy Systems*, Vol. 17, No. 2, pp.292–302.

Yuen, K. and Fung, K. (2014) 'Combining compound linguistic ordinal scale and cognitive pairwise comparison in the rectified fuzzy TOPSIS method for group decision making', *Fuzzy Optimization and Decision Making*, Vol. 13, No. 1, pp.105–130.

Zadeh, L.A., Fu, K-S. and Tanaka, K. (2014) 'Fuzzy sets and their applications to cognitive and decision processes', *Proceedings of the US-Japan Seminar on Fuzzy Sets and Their Applications*, University of California, Berkeley, California, 1–4 July 1974, Academic Press.

Zeng, X-T., Li, D-F. and Yu, G-F. (2014) 'A Value and ambiguity-based ranking method of trapezoidal intuitionistic fuzzy numbers and application to decision making', *The Scientific World Journal*, 22 July, doi:10.1155/2014/560582.

Zientara, P., Kujawski, L. and Bohdanowicz-Godfrey, P. (2015) 'Corporate social responsibility and employee attitudes: evidence from a study of Polish hotel employees', *Journal of Sustainable Tourism*, Vol. 23, No. 6, pp.859–880.